

Name: _____

Class: _____

Date: _____

ID: A

Chemistry Chapter 17 Practice Quiz 3 — Keq Calculations

Show ALL work on every calculation. Write what you know → formula → plug in → evaluate.

Multiple Choice

Identify the choice that best completes the statement or answers the question.

- ___ 1. For the reaction $2\text{NO(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$, the correct Keq expression is
- a. $\text{Keq} = [\text{NO}_2]^2 / ([\text{NO}]^2[\text{O}_2])$ b. $\text{Keq} = [\text{NO}]^2[\text{O}_2] / [\text{NO}_2]^2$
c. $\text{Keq} = [\text{NO}_2] / [\text{NO}][\text{O}_2]$ d. $\text{Keq} = 2[\text{NO}_2] / 2[\text{NO}][\text{O}_2]$
- ___ 2. For the reaction $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$, the Keq expression is
- a. $\text{Keq} = [\text{CaO}][\text{CO}_2] / [\text{CaCO}_3]$ b. $\text{Keq} = [\text{CO}_2]$
c. $\text{Keq} = [\text{CaCO}_3] / [\text{CaO}][\text{CO}_2]$ d. $\text{Keq} = 1 / [\text{CO}_2]$
- ___ 3. For the reaction $\text{Mg(s)} + 2\text{HCl(aq)} \rightleftharpoons \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$, at equilibrium $[\text{HCl}] = 0.051 \text{ M}$, $[\text{MgCl}_2] = 0.513 \text{ M}$, $[\text{H}_2] = 0.411 \text{ M}$. The Keq is approximately
- a. 4.1 b. 41
c. 81 d. 0.081
- ___ 4. If $\text{Keq} = [\text{NO}_2]^2 / ([\text{NO}]^2[\text{O}_2])$ and $[\text{NO}] = 0.897 \text{ M}$, $[\text{O}_2] = 0.254 \text{ M}$, $[\text{NO}_2] = 0.0329 \text{ M}$, then Keq is approximately
- a. 5.30×10^{-3} b. 5.30×10^{-1}
c. 0.530 d. 53.0
- ___ 5. A student writes $\text{Keq} = [\text{NH}_3] / [\text{N}_2][\text{H}_2]$ for $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$. What error did the student make?
- a. Products should be in the denominator b. The coefficients (2 and 3) must be used as exponents
c. N_2 and H_2 should not be included since they are gases d. There is no error

True / False

Indicate whether the statement is true or false.

- ___ 6. In the Keq expression, the exponents on each concentration term equal the coefficients in the balanced equation.
- ___ 7. Pure solids appear in the Keq expression with a concentration value of 1.
- ___ 8. A large Keq value means the denominator (reactants) is larger than the numerator (products).
- ___ 9. The Keq value must always be determined experimentally; it cannot be calculated from the equation alone.
- ___ 10. For the reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$, if $[\text{A}] = 2$, $[\text{B}] = 2$, $[\text{C}] = 1$, $[\text{D}] = 1$, then $\text{Keq} = 0.25$.

- ___ 11. Doubling all concentrations in a reaction at equilibrium will double the value of K_{eq} .
- ___ 12. Aqueous ions ARE included in the K_{eq} expression.
- ___ 13. For homogeneous equilibria, all substances in the K_{eq} expression are in the same phase.
- ___ 14. K_{eq} has no units — it is a dimensionless number.
- ___ 15. If $[products] > [reactants]$ at equilibrium, then $K_{eq} > 1$.

Calculations — Show All Work

16. Calculate K_{eq} for the formation of ammonia: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $[N_2] = 0.045\text{ M}$, $[H_2] = 0.055\text{ M}$, $[NH_3] = 52\text{ M}$ (Example 17-3 from your textbook)

17. Calculate K_{eq} for the reaction: $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $[NO] = 0.897\text{ M}$, $[O_2] = 0.254\text{ M}$, $[NO_2] = 0.0329\text{ M}$

18. Write the K_{eq} expression AND calculate K_{eq} for the reaction of solid magnesium with hydrochloric acid: $Mg(s) + 2HCl(aq) \rightleftharpoons MgCl_2(aq) + H_2(g)$ $[HCl] = 0.051\text{ M}$, $[MgCl_2] = 0.513\text{ M}$, $[H_2] = 0.411\text{ M}$

Essay

Answer TWO of the following. You may answer all 3 for possible bonus.

19. Explain step by step how you would calculate K_{eq} given a balanced chemical equation and equilibrium concentrations. Use the formation of ammonia as your worked example, showing every step clearly.

20. Explain the rule for writing K_{eq} expressions from balanced equations. What happens to solids and liquids? What do the exponents represent? Why does the numerical value of K_{eq} have to be determined experimentally?

21. A student calculates $K_{eq} = 8.2 \times 10^{-10}$ for a reaction. Another student calculates $K_{eq} = 4.5 \times 10^5$ for a different reaction. Compare these two reactions in terms of: (a) which is more product-favored, (b) where equilibrium lies for each, and (c) what happens to [products] vs [reactants] at equilibrium for each.